

Cramer's Rule.

1. A total of 8,500 taka was invested in three interest earning accounts. The interest rates were 2%, 3% and 6% if the total simple interest for one year was 380 taka and the amount invested at 6% was equal to the sum of the amounts in the other two accounts, then how much was invested in each account? (Use Cramer's rule).

Let the amounts invested in the three accounts be Tk. x, Tk. y and Tk. z

Interest for the three accounts are $(2/100)x$, $(3/100)y$ and $(6/100)z$

According to the problem,

$$x + y + z = 8500 \dots\dots\dots (1)$$

$$(2/100)x + (3/100)y + (6/100)z = 380 \text{ (or) multiplying by 100,}$$

$$2x + 3y + 6z = 38000 \dots\dots\dots (2)$$

$$z = x + y \text{ or } x + y - z = 0 \dots\dots\dots (3)$$

$$\text{Now, } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 6 \\ 1 & 1 & -1 \end{vmatrix} = 1(-3-6) - 1(-2-6) + 1(2-3) \\ = -9 + 8 - 1 \\ = -2 \neq 0$$

So there exists a unique solution to the system (1), (2) and (3)

$$\Delta_x = \begin{vmatrix} 8500 & 1 & 1 \\ 38000 & 3 & 6 \\ 0 & 1 & -1 \end{vmatrix} = 8500(-3-6) - 1(-38000) + 1(38000) \\ = -76500 + 76000 \\ = -500$$

$$\Delta_y = \begin{vmatrix} 1 & 8500 & 1 \\ 2 & 38000 & 6 \\ 1 & 0 & -1 \end{vmatrix} = 1(-38000) - 8500(-2-6) + 1(-38000) \\ = -38000 + 68000 - 38000 \\ = -8000$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 8500 \\ 2 & 3 & 38000 \\ 1 & 1 & 0 \end{vmatrix} = 1(-38000) - 1(-38000) + 8500(2-3) \\ = -38000 + 38000 - 8500 \\ = -8500$$

So by Cramer's rule,

$$x = \frac{\Delta_x}{\Delta} = \frac{-500}{-2} = 250, \quad y = \frac{\Delta_y}{\Delta} = \frac{-8000}{-2} = 4000, \quad z = \frac{\Delta_z}{\Delta} = \frac{-8500}{-2} = 4250$$

Thus the amount invested at 2% is Tk. 250, at 3% is Tk. 4000 and at 6% is Tk. 4250.

2. In a market survey three commodities A, B and C were considered. In finding out the index number some fixed weights were assigned to the three varieties in each of the commodities. The table below provides the information regarding the consumption of three commodities according to the three varieties and also the total weight received by the commodity. Use Cramer's rules.

Commodity Variety	Variety			Total weight
	I	II	III	
A	1	2	3	11
B	2	4	5	21
C	3	5	6	27

Solⁿ:

Let the weight assigned to the three varieties be Rs. x , Rs. y and Rs. z respectively.

By the given data,

$$x + 2y + 3z = 11$$

$$2x + 4y + 5z = 21$$

$$3x + 5y + 6z = 27$$

$$\Delta = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{vmatrix} = 1 \begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} - 2 \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} + 3 \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix}$$

$$= 1(24 - 25) - 2(12 - 15) + 3(10 - 12)$$

$$= 1(-1) - 2(-3) + 3(-2)$$

$$= -1 + 6 - 6 = -1 \neq 0.$$

Since $\Delta \neq 0$, the system is consistent with unique solution and Cramer's rule can be applied.

$$\Delta x = \begin{vmatrix} 11 & 2 & 3 \\ 21 & 4 & 5 \\ 27 & 5 & 6 \end{vmatrix}$$

$$= 11 \begin{vmatrix} 4 & 5 \\ 5 & 6 \end{vmatrix} - 2 \begin{vmatrix} 21 & 5 \\ 27 & 6 \end{vmatrix} + 3 \begin{vmatrix} 21 & 4 \\ 27 & 5 \end{vmatrix}$$

$$= 11(24 - 25) - 2(126 - 135) + 3(105 - 108)$$

$$= 11(-1) - 2(-9) + 3(-3)$$

$$= -11 + 18 - 9$$

$$= -2$$

$$\Delta y = \begin{vmatrix} 1 & 11 & 3 \\ 2 & 21 & 5 \\ 3 & 27 & 6 \end{vmatrix}$$

$$= \begin{vmatrix} 21 & 5 \\ 27 & 6 \end{vmatrix} - 11 \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} + 3 \begin{vmatrix} 2 & 21 \\ 3 & 27 \end{vmatrix}$$

$$= 1(126 - 135) - 11(12 - 15) + 3(54 - 63)$$

$$= -9 - 11(-3) + 3(-9)$$

$$= -9 + 33 - 27$$

$$= -3$$

$$\Delta z = \begin{vmatrix} 1 & 2 & 11 \\ 2 & 4 & 21 \\ 3 & 5 & 27 \end{vmatrix} = 1 \begin{vmatrix} 4 & 21 \\ 5 & 27 \end{vmatrix} - 2 \begin{vmatrix} 2 & 21 \\ 3 & 27 \end{vmatrix} + 11 \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix}$$

$$= 1(108 - 105) - 2(54 - 63) + 11(10 - 12)$$

$$= 1(3) - 2(-9) + 11(-2)$$

$$= 3 + 18 - 22$$

$$= -1$$

$$x = \frac{\Delta x}{\Delta} = \frac{-2}{-1} = 2$$

$$y = \frac{\Delta y}{\Delta} = \frac{-3}{-1} = 3$$

$$\text{and } z = \frac{\Delta z}{\Delta} = \frac{-1}{-1} = 1$$

Hence, the weights assigned to the three varieties are 2, 3 and 1 respectively.

3. Three masses are suspended vertically by a series of identical springs where mass 1 is at the top and mass 3 is at the bottom. If $g=9.8\text{m/s}^2$, $m_1=2\text{kg}$, $m_2=3\text{kg}$, $m_3=2.5\text{kg}$ and $k's=10\text{kg/s}$, solve for the displacements x . Apply Cramer's rule and find the solution.

Consider that the three masses suspended vertically by a series of identical spring. Thus, the steady-state force is balanced between three masses are represented by,

$$\begin{aligned}m_1g - kx_1 + k(x_2 - x_1) &= 0 \\m_2g - k(x_2 - x_1) + k(x_3 - x_2) &= 0 \\m_3g - k(x_3 - x_2) &= 0\end{aligned}$$

Solve for standard form of

$$\begin{aligned}m_1g - kx_1 + k(x_2 - x_1) &= 0 \\kx_1 - k(x_2 - x_1) &= m_1g \\kx_1 - kx_2 + kx_1 &= m_1g \\2kx_1 - kx_2 &= m_1g\end{aligned}$$

Solve for the standard form of

$$\begin{aligned}m_2g - k(x_2 - x_1) + k(x_3 - x_2) &= 0 \\k(x_2 - x_1) - k(x_3 - x_2) &= m_2g \\kx_2 - kx_1 - kx_3 + kx_2 &= m_2g \\-kx_1 + 2kx_2 - kx_3 &= m_2g\end{aligned}$$

Solve for standard form of

$$\begin{aligned}m_3g - k(x_3 - x_2) &= 0 \\-kx_2 + kx_3 &= m_3g\end{aligned}$$

Now write the term in form of,

$$\begin{aligned}2kx_1 - kx_2 + 0 \cdot x_3 &= m_1g \\-kx_1 + 2kx_2 - kx_3 &= m_2g \\0 \cdot x_1 - kx_2 + kx_3 &= m_3g\end{aligned}$$

Now write the term in form of,

$$\begin{aligned} 20x_1 - 10x_2 + 0 \cdot x_3 &= 19.62 \\ -10x_1 + 20x_2 - 10x_3 &= 29.43 \\ 0 \cdot x_1 - 10x_2 + 10x_3 &= 24.525 \end{aligned}$$

Now, we apply Cramer's rule and find the solution:

$$X_1 = 7.3575$$

$$X_2 = 12.753$$

$$X_3 = 15.2055$$

4. A total of ₹8,600 was invested in two accounts. One account earned $4\frac{3}{4}\%$ annual interest and the other earned $6\frac{1}{2}\%$ annual interest. If the total interest for one year was ₹431.25, how much was invested in each account? (Use Cramer's Rule).

Solⁿ:

Let the amount invested in the two accounts be Rs x and Rs. y respectively

$$\text{By the given data, } x + y = 8600 \quad \text{----- (1)}$$

$$4\frac{3}{4} \times \frac{x}{100} + 6\frac{1}{2} \times \frac{y}{100} = 431.25$$

$$[\therefore \text{interest} = \frac{\text{PNR}}{100}]$$

$$\Rightarrow \frac{19x}{400} + \frac{13y}{200} = 431.25$$

$$\Rightarrow \frac{19x + 26y}{400} = 431.25$$

$$19x + 26y = 172500 \quad \text{----- (2)}$$

$$\Delta = \begin{vmatrix} 1 & 1 \\ 19 & 26 \end{vmatrix} = 1(26) - 1(19)$$

$$= 26 - 19 = 7$$

$$\Delta x = \begin{vmatrix} 8600 & 1 \\ 172500 & 26 \end{vmatrix} = 8600(26) - 1(172500)$$

$$= 223600 - 172500 = 51100$$

$$\Delta y = \begin{vmatrix} 1 & 8600 \\ 19 & 172500 \end{vmatrix} = 1(172500) - 19(8600)$$

$$= 172500 - 163400 = 9100$$

$$\therefore x = \frac{\Delta x}{\Delta} = \frac{51100}{7} = 7300$$

$$y = \frac{\Delta y}{\Delta} = \frac{9100}{7} = 1300$$

$\therefore$ Investment in the interest of $4\frac{3}{4}\%$ account is Rs. 7300 and investment in the rate of $6\frac{1}{2}\%$ account is Rs. 1300.

1. Consider an online store selling T-shirts (T) and Hoodies (H). The prices are as follows:

✓ A T-shirt costs \$10.

✓ A Hoodie costs \$20.

A customer buys three items (either T-shirts or Hoodies) for a total cost of \$50. How many T-shirts (x) and Hoodies(y) did the customer buy? Solve Using Gauss-Jordan method.

Solve using Gauss-elimination method.

The objective is to find the values of x and y that satisfy the following system of linear equations:

$$10x + 20y = 50 \text{ (Total cost)}$$

$$x + y = 3 \text{ (Total Number of items)}$$

Converting given equations into matrix form

$$R_1 \leftarrow R_1 / 10$$

$$R_2 = R_1 - R_2$$

$$R_1 = R_1 - R_2 \times 2$$

From the above augmented matrix we find:

$$y = 2$$

and

$$x = 1$$

so, the customer bought x=1 Shirts and y=2 Hoodie.

2. A total of ₹8,600 was invested in two accounts. One account earned $4\frac{3}{4}\%$ annual interest and the other earned $6\frac{1}{2}\%$ annual interest. If the total interest for one year was ₹431.25, how much was invested in each account? (Use Gauss-Jordan Method).

Solⁿ:

Let the amount invested in the two accounts be Rs x and Rs. y respectively

By the given data, $x + y = 8600$ ----- (1)

$$4\frac{3}{4} \times \frac{x}{100} + 6\frac{1}{2} \times \frac{y}{100} = 431.25$$

$$[\therefore \text{interest} = \frac{\text{PNR}}{100}]$$

$$\Rightarrow \frac{19x}{400} + \frac{13y}{200} = 431.25$$

$$\Rightarrow \frac{19x + 26y}{400} = 431.25$$

$$19x + 26y = 172500 \text{ ----- (2)}$$

Converting given equations into matrix form

$$\left[\begin{array}{cc|c} 1 & 1 & 8600 \\ 19 & 26 & 172500 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 19 \times R_1$$

$$= \left[\begin{array}{cc|c} 1 & 1 & 8600 \\ 0 & 7 & 9100 \end{array} \right]$$

$$R_2 \leftarrow R_2 \div 7$$

$$= \left[\begin{array}{cc|c} 1 & 1 & 8600 \\ 0 & 1 & 1300 \end{array} \right]$$

$$R_1 \leftarrow R_1 - R_2$$

$$= \left[\begin{array}{cc|c} 1 & 0 & 7300 \\ 0 & 1 & 1300 \end{array} \right]$$

i. e.

$$x = 7300$$

$$y = 1300$$

Solution By Gauss jordan elimination method

$$x = 7300 \text{ and } y = 1300$$

3. An investor has \$10,000 to invest in two types of financial instruments: stocks and bonds. The expected return from stocks is 8%, and from bonds is 5%. The investor wants the total return to be \$700. Find the amount invested in each type Using Gauss-Jordan method.

Solve using Gauss-Jordan method.

from the Given problem we find two equations:-

$$0.08x + 0.05y = 700 \text{-----(i)}$$

$$x + y = 10000 \text{-----(ii)}$$

here, x=stocks and y=bonds.

Converting given equations into matrix form

$$R_1 \leftarrow R_1 / 0.08$$

$$R_2 \leftarrow R_2 - R_1$$

$$R_2 \leftarrow R_2 / .375$$

$$R_1 \leftarrow R_1 - .625xR_2$$

So, $x = 6666.667$ and $y = 3333.3333$.

Hence amount of stocks invested 6666 and amount of bonds invested 3333.

4. A commodity was produced by using 3 units of labor and 2 units of capital, the total cost is 62 Tk. If the commodity had been produced by using 4 units of labor and one unit of capital, the cost is 56 Tk. What is the cost per unit of labor and capital? (Use Gauss-Jordan Method).

Solⁿ:

Let, Tk. x represents the cost per unit labor and

Tk. y represents the cost per unit of capital

From above information we find

$$3x + 2y = 62 \text{-----}\{1\}$$

$$4x + y = 56 \text{-----}\{2\}$$

Converting given equations into matrix form

$$\left[\begin{array}{cc|c} 3 & 2 & 62 \\ 4 & 1 & 56 \end{array} \right]$$

$$R_1 \leftarrow R_1 \div 3$$

$$= \left[\begin{array}{cc|c} 1 & 0.6667 & 20.6667 \\ 4 & 1 & 56 \end{array} \right]$$

$$R_2 \leftarrow R_2 - 4 \times R_1$$

$$= \left[\begin{array}{cc|c} 1 & 0.6667 & 20.6667 \\ 0 & -1.6667 & -26.6667 \end{array} \right]$$

$$R_2 \leftarrow R_2 \times -0.6$$

$$= \left[\begin{array}{cc|c} 1 & 0.6667 & 20.6667 \\ 0 & 1 & 16 \end{array} \right]$$

$$R_1 \leftarrow R_1 - 0.6667 \times R_2$$

$$= \left[\begin{array}{cc|c} 1 & 0 & 10 \\ 0 & 1 & 16 \end{array} \right]$$

And now, you need to solve all Cramer's Rule problem by Gauss-Jordan Method and Vice-versa.

One set question from this type.